MAIS PAC

IPhO 02.08: Circuits, Capacitors and Inductors · Lecture + Problems

Circuits, Capacitors and Inductors

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

1. What makes a circuit hard
2. Kirchhoff's rules, done properly
3. The RC circuit: exponentials from a step
4. The RL circuit: inertia for current
5. LC oscillations and resonance
6. Problems: reason, calculate, check
7. Further practice and provenance
8. Where to learn more

1. What makes a circuit hard

In electrostatics, charges sit still. In a circuit, charges are pushed round a closed path, and the difficulty is not the arithmetic but the **bookkeeping**: which quantity is shared, and which is divided up.

Three ideas organise everything in this topic:

- **Kirchhoff's rules.** Charge is conserved at every junction, and energy is conserved round every loop. Both stay true whether the elements are linear or not.
- **Components store or dissipate.** A resistor turns electrical energy into heat and has no memory. A capacitor stores energy in an electric field, an inductor in a magnetic field, and both have memory: their state cannot change instantaneously.
- **Time and frequency are two views of the same circuit.** A step input is described by exponentials with a time constant; a sinusoidal input is described by impedances and phasors. Olympiad problems use one or the other, and choosing well halves the work.

2. Kirchhoff's rules, done properly

Label every branch current with a direction. The direction you guess does not matter: a wrong guess simply produces a negative number, which is a correct answer, not an error.

- **Junction rule:** Σ I_({in}) = Σ I_({out}). Charge cannot pile up at a point in a steady circuit.
- **Loop rule:** Σ Δ V = 0 around any closed loop. Going **with** a current through R drops the potential by IR; going through a source from - to + raises it by ε.

The two rules are not independent of the element laws — they only supply the equations, while V = IR, Q = CV and ε = -L,dI/dt describe each element. A network with b branches and n junctions needs b independent equations: n-1 junction equations and b-n+1 loop equations.

Two shortcuts worth practising until they are automatic:

- **Series/parallel reduction.** Elements in series share the same current and add their voltages (so R add, (1)/(C) add); elements in parallel share the same voltage and add their currents ((1)/(R) add, C add).
- **Superposition and symmetry.** In a linear circuit you may solve one source at a time and add the answers. If a network is symmetric, some branches must carry no current at all — often the fastest route to the answer.

3. The RC circuit: exponentials from a step

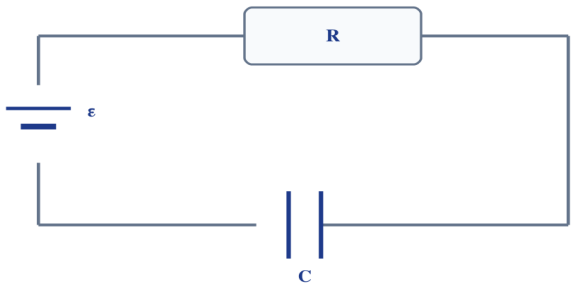
Put a battery, a resistor and a capacitor in series. The loop rule reads

$$\varepsilon = IR + (q)/(C),\ I = (dq)/(dt), \tag{1}$$

which is a first-order differential equation. Its solution from an empty capacitor is

$$q(t) = C\varepsilon(1 - e^{(-t/\tau)}),\ I(t) = (\varepsilon)/(R)e^{(-t/\tau)},\ \tau = RC. \tag{2}$$

An RC circuit: the resistor sets the current, the capacitor stores the charge



ε = IR + q/C ⟹ q = Cε(1 − e^(−t/RC)); the current starts at ε/R and dies away

Figure 2. At t = 0 the capacitor is empty, so it behaves like a plain wire; after a long time it is full and behaves like a break. Those two limits answer most RC questions without algebra.

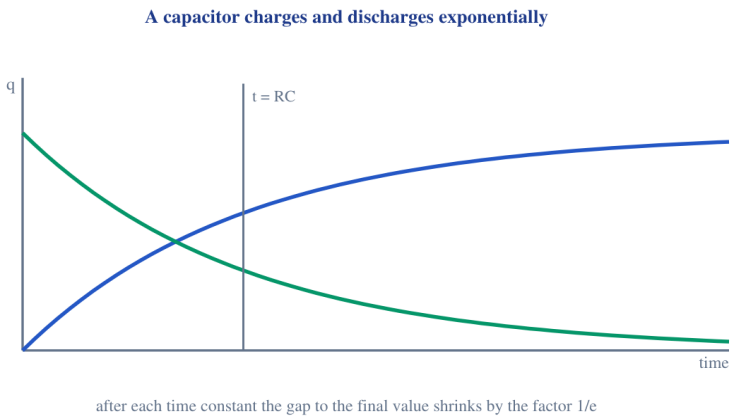


Figure 1. The two curves are mirror imag

you have covered 63% of it.
After one time constant the capacitor has 63% of its final charge and the current has fallen to 37% of its initial value. After five, it is "finished" for any practical purpose.

Three facts carry most of the marks:

- At t=0 an uncharged capacitor behaves like a **wire** (it can carry any current, with no voltage across it).
- After a long time a capacitor behaves like a **break** (no steady current, so it carries the full source voltage).
- The energy accounting is surprising: the battery supplies Cε² while the capacitor stores only (1)/(2)Cε². The other half is dissipated in the resistor **whatever** the value of R. With R→0 the power rises and the time shrinks, and the product stays fixed.

Discharging works the same way with the source removed: q = Q_0e^(-t/τ), and the stored energy is lost to the resistor.

4. The RL circuit: inertia for current

Replace the capacitor with an inductor and the equation becomes

$$\varepsilon = IR + L(dI)/(dt)\ I(t) = (\varepsilon)/(R)(1 - e^{(-t/\tau)}),\ \tau = (L)/(R). \quad (3)$$

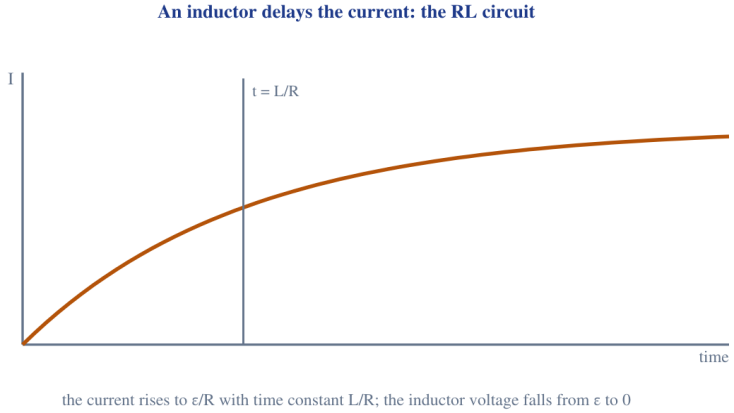


Figure 3. Compare with the

capacitor: there the *voltage* rises with τ = RC, here the *current* rises with τ = L/R. An inductor is inertia for current.
Now it is the **current** that cannot jump, and the roles of the two limits swap: at t=0 the inductor behaves like a **break** (it takes the whole source voltage, L,dI/dt = ε, while I=0), and after a long time it behaves like a **wire**.

That behaviour is the origin of the spark when you unplug an inductive device: interrupting the current forces dI/dt to be huge and negative, so L,dI/dt becomes a large voltage that can break down the air. It is also why a large inductor must never be disconnected abruptly.

5. LC oscillations and resonance

Join a charged capacitor to an inductor with no resistance. The capacitor discharges through the inductor, the inductor's back-emf keeps the current flowing until the capacitor is charged the other way, and the process repeats:

$$L(d^2q)/(dt^2) + (q)/(C) = 0 \; q = Q_0\cos(\omega_0 t), \; \omega_0 = (1)/(\sqrt{(LC)}). \tag{4}$$

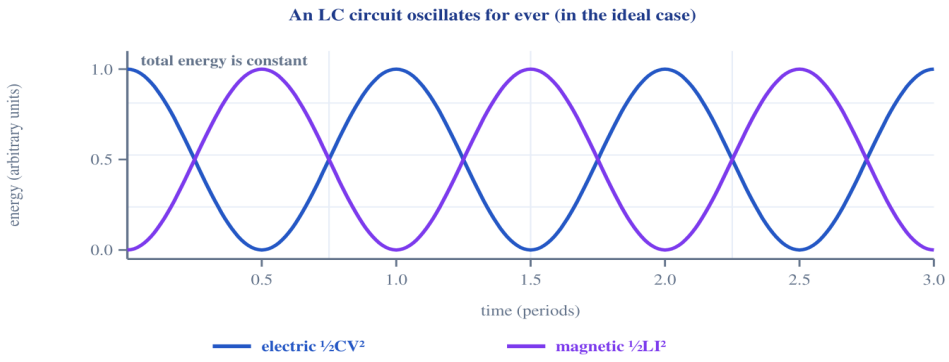


Figure. Energy shuttles between the capacitor and the inductor with angular frequency 1/sqrt(LC).

The energy shuttles between the electric field of the capacitor and the magnetic field of the inductor, and in the ideal case the total stays constant. This is a harmonic oscillator in a different costume: q plays the role of displacement, I of velocity, 1/C of spring constant and L of mass.

Add a resistance and the oscillation decays with time constant 2L/R; the circuit is underdamped, critically damped or overdamped according to whether R^2 is smaller than, equal to, or larger than 4L/C.

For a **sinusoidally driven** series RLC circuit the steady state is best found with impedances:

$$X_L = \omega L, \; X_C = (1)/(\omega C), \; Z = \sqrt{(R^2 + (X_L - X_C)^2)}, \; I_0 = (\varepsilon_0)/(Z). \tag{5}$$

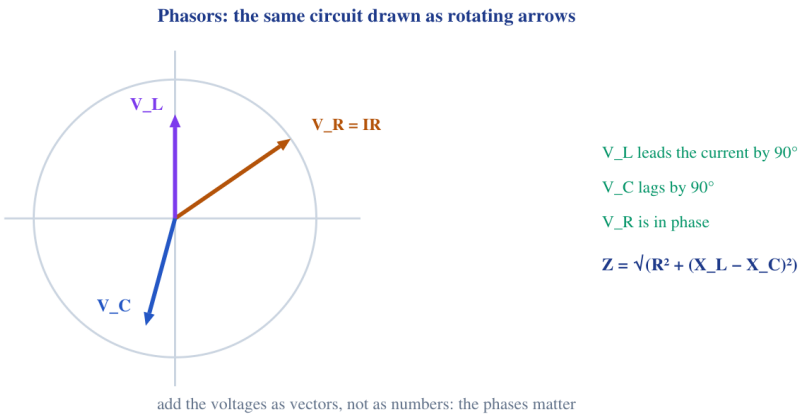


Figure 5. At resonance X_L = X_C, the two reactive arrows cancel, and the current is limited only by R — which is why a high-Q circuit can carry

enormous currents.

At the resonant frequency ω_0 = 1/√(LC) the two reactances cancel, Z = R, and the current peaks. The sharpness of that peak is measured by the quality factor

$$Q = (\omega_0 L)/(R) = (1)/(R)\sqrt{((L)/(C))}, \{\text{bandwidth }\} \Delta\omega = (\omega_0)/(Q). \quad (6)$$

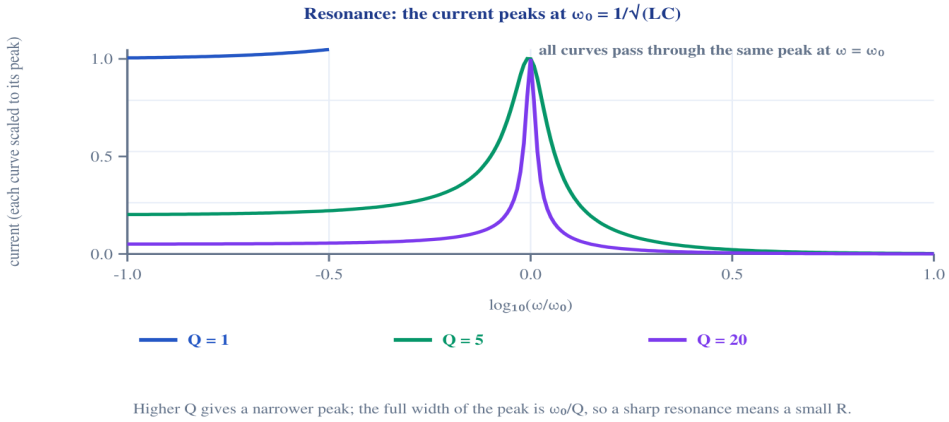


Figure. Higher Q gives a narrower peak; the full width of the peak is ω■/Q, so a sharp resonance means a small R.

At resonance the voltages across the capacitor and the inductor are each Q times the supply voltage and exactly opposite in phase. Nothing is violated: their **sum** is zero at every instant, and the energy sloshing between them comes from the source only through the small resistive loss.

Interactive experiment: Resonance in a series RLC circuit — open the online handout to use it.

The circuit checklist. Transient or steady? (step → time constant; sine → impedance) → what is shared, current or voltage? → apply the junction and loop rules → check the two limits t=0 and t→∞ (capacitor: wire then break; inductor: break then wire) → for resonance, compute ω_0, Q and the bandwidth, and remember that reactive voltages can exceed the supply.

6. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a worked solution with a consistency check.

[1] Problem 1 — Charging through a resistor. A 12 {V} battery charges a 100 μ{F} capacitor through a 2.0 {kΩ} resistor. (a) Find the time constant and the initial current. (b) Find the charge and the current after 0.30 {s}. (c) Find the energy stored in the capacitor at that time and the total energy supplied by the battery so far.

Worked solution →

[1] Problem 2 — Discharging through two paths. A 470 μ{F} capacitor is charged to 24 {V} and then discharged through a 1.5 {kΩ} resistor. (a) Find the initial stored energy. (b) Find the time for the voltage to fall to 6.0 {V}. (c) Show that the total energy delivered to the resistor equals the initial stored energy, whatever the resistance.

Worked solution →

[2] Problem 3 — An inductor switched on. A coil of inductance 0.25 {H} and resistance 5.0 Ω is connected to a 6.0 {V} supply of negligible internal resistance. (a) Find the final current. (b) Find the time constant and the current after 30 {ms}. (c) Find the voltage across the inductor at t=0 and at t=30 {ms}, and explain the limit t→∞.

Worked solution →

[2] Problem 4 — LC oscillation. A 2.0 μ{F} capacitor is charged to 50 {V} and connected across a 20 {mH} inductor. (a) Find the oscillation frequency and period. (b) Find the maximum current. (c) Find the energy in the magnetic field when the capacitor voltage has fallen to 25 {V}, and check that the total is conserved.

Worked solution →

[2] Problem 5 — A driven series RLC circuit. A series circuit has R = 10 Ω, L = 50 {mH} and C = 20 μ{F}, driven by a 10 {V} (rms) source of variable frequency. (a) Find the resonant frequency. (b) Find the rms current at resonance. (c) Find the rms voltage across the capacitor at resonance and explain how it can exceed the supply.

Worked solution →

[2] Problem 6 — Quality factor and bandwidth. For the circuit of Problem 5, (a) find the quality factor Q and the bandwidth. (b) Find the two frequencies at which the current is $1/\sqrt{2}$ of its peak value. (c) Explain what happens to the peak current as R is reduced towards zero and what limits the effect in practice.

Worked solution →

[2] Problem 7 — A capacitor network. Three capacitors C_1 = 4.0 μ{F}, C_2 = 6.0 μ{F} and C_3 = 12 μ{F} are connected across a 24 {V} supply with C_1 in series with the parallel pair C_2 C_3. (a) Find the equivalent capacitance. (b) Find the charge and voltage on each capacitor. (c) Find the total energy stored and compare it with the energy if all three were in parallel.

Worked solution →

[3] Problem 8 — Two capacitors sharing charge. A 10 μ{F} capacitor charged to 100 {V} is connected to an uncharged 20 μ{F} capacitor. (a) Find the final voltage and the final charges. (b) Find the energy before and after, and account for the difference. (c) Explain why the loss does not depend on the resistance of the connecting wire.

Worked solution →

[3] Problem 9 — An RC timing circuit. In an RC circuit with ε = 9.0 {V}, R = 50 {kΩ} and C = 22 μ{F}, a lamp switches on when the capacitor voltage reaches 4.0 {V}. (a) Find the delay. (b) Find the current at that instant. (c) Explain what happens to the delay if the capacitor is replaced by one of twice the capacitance, and why such circuits are used as timers.

Worked solution →

[3] Problem 10 — A capacitor, a switch and two resistors. A 12 {V} battery, a 30 {kΩ} resistor R_1, a 60 {kΩ} resistor R_2 and a 5.0 μ{F} capacitor are connected so that R_1 and R_2 are in series across the battery and the capacitor is connected across R_2. The capacitor starts uncharged. (a) Find the final voltage across the capacitor and the steady currents. (b) Find the time constant of the charging. (c) Find the time at which the capacitor voltage is 6.0 {V}, and explain why the relevant resistance is R_1R_2/(R_1+R_2) and not R_1+R_2.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

> The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.
>
> MIT OpenCourseWare 8.02, Electricity and Magnetism
> Full lecture notes and problem sets for fields, circuits and induction.